CineLog
Phase 2 Database Project Report

Group 10

Names	Roll numbers
Hanzalah Sohail Faraz	**26100028**
Raahem Azam	**26020347**
Muhammad Hashaam Tauseef	**26020105**
Azaan Imran	**26100281**

1. Overview of Database Design

The CineLog database is designed as a scalable, partitioned, analytics-ready movie logging system supporting user activity, movie metadata, genres, reviews, lists, watchlists, and social connections. The system integrates a large cleaned Kaggle movie dataset (~940k movies) and generates substantial synthetic data (1M reviews, 1000 users, 20k list items) to simulate a real production environment. Key design features include partitioning on the Reviews table, many-to-many modeling of Movie-Genre relationships, indexing for fast lookups, and automated logic implemented through triggers, views, stored procedures, and functions.

2. Schema Summary

Core Tables

- **Users**: 1000 generated users with profile details.
- **Movies**: 940k movie entries imported from cleaned CSV.
- **Genres**: 20 unique genres extracted from the dataset.
- **MovieGenres**: 1.35M mappings (many-to-many).
- **Reviews (Partitioned)**: 1,000,000 partitioned review entries.

- **Watchlist**: Movie pairs of 5,000 users.
- **Follow**: 2000 unique user-follow relationships.
- **Lists**: 3000 auto-generated lists (Eid Picks, Favorites, Christmas Picks).
- **ListItem**: 20,000 list–movie items.

Partitioning

- **Partition Function:** partition_ReviewDate defined on DATETIME.
- **Partition Scheme:** partition_ReviewDate mapped to PRIMARY.
- Reviews table created ON this partition scheme using ReviewDate.
- Clustered index aligned with the partition scheme for high performance.

3. Features Implemented

A. Triggers

- **trigger_UpdateMovieRating:** Automatically recalculates VoteAverage and VoteCount whenever reviews are inserted, updated, or deleted.
- **trgger_CleanupWatchlist:** When a user posts a review, the movie is automatically removed from their Watchlist.
- **trigger_CheckReviewDate:** INSTEAD OF INSERT trigger that blocks inserting reviews with a future date.

B. Stored Procedures

- **proced_AddReview**: Securely inserts a user review with current timestamp.
- **proced_FollowUser**: Adds a follow relationship if not already present.
- **proced_GetSuperUsers**: Finds users with above-average review counts.
- **proced_FindFriendConnections**: Recursive 3 level friend of friend traversal.
- **Proced_GetMoviesByGenre**: Returns movies filtered by a given genre.

- **proced_GetUserActivity**: Merges Reviews + Watchlist + List Items into a single activity feed.
- **proced_GetTrendingMovies**: Ranks movies by review volume in last N days.
- **proced_RecommendMovies**: Simple recommender using top genres of a user.
- **proced_GetMovieDetails**: Returns movie info, genres, and 20 recent reviews.

C. Functions

- **fn_GetUserWatchCount**: Returns number of reviews posted by a user.
- **fn_FormatRuntime**: Converts runtime minutes to "Xh Ym" text.
  rage fo
- **fn_AvgRatingByGenre**: Returns the average VoteAverage for a given genre.

D. Views

- **vw_TopRatedMovies**: Top 100 highest-rated movies (VoteCount > 50).
- **vw_PublicActivityFeed**: Latest 50 public reviews.
- **vw_MostActiveUsers**: Top 50 users by review count.
- **vw_PopularGenres**: Top 20 most common genres across all movies.

E. Indexing

- **IX_Movies_Title**: speeds up title searching.
- **IX_Reviews_UserID**: accelerates profile loading & activity queries.

4. Data Import Pipeline

- CSV imported from /var/opt/mssql/movies_utf8.csv into Staging_Movies.
- Null / empty titles removed.
- Cleaned fields mapped into the Movies table.
- Genres extracted using STRING_SPLIT with whitespace trimming.

- Invalid genre values filtered out with NOT LIKE '%[^a-zA-Z]%'.
- Final: 20 genres inserted, 1.35M Movie-Genre pairs generated.

5. Data Generation

- **Users:** 1000 auto-generated.
- **Reviews:** 1 million generated via cross-join of system tables.
- **Follows:** 2000 unique followers followed pairs.
- **Watchlist:** 5000 random movie additions.
- **Lists:** 3 lists/user → 3000 lists.
- **ListItem:** 20,000 random list–movie mappings.

6. Final Row Counts

Table	Rows
Movies	**940,697**
Reviews	**1,000,000**
Users	**1,000**
Genres	**20**
MovieGenres	**1,353,358**
Lists	**3,000**
ListItem	**20,000**
WatchList	**5000**
Follows	**2000**